Solving Puzzles with SAT Solvers

Stop writing algorithms. Start stating the rules.

Meet Z3

What on Earth is Z3?

- A SAT solver (and more!) from Microsoft Research
- Solves boolean satisfiability problems: can we make these constraints true?
- You describe **what** must be true
- Z3 works out **how** — or proves it's impossible
- *(Bonus: Z3 also handles integers, reals, arrays, etc. — it's a full SMT solver! You will learn this later.)

You, the wizard

"Exactly one room hides the Lady. Exactly one sign on the doors is telling the truth."

Z3, the genie

"Then she is behind door one." ✓

```
[R1 = True, R2 = False, R3 = False,
...]
```

Declarative, not procedural

Normal programming

- You write the *steps*
- **You** are the algorithm designer
- Loops, conditionals, backtracking...
- Bug in your search $\Rightarrow$ wrong answer

Z3 programming

- You write the *constraints*
- **Z3** is the algorithm designer
- You just state the rules of the game
- Rules right $\Rightarrow$ answer right

You bring the rules.

Z3 brings the answer.

Under the Hood: SAT Solving Algorithms

DPLL Algorithm

You've learned this in class!

- Backtracking search
- Unit propagation
- Pure literal elimination

Modern SAT Solvers (CDCL)

What Z3 actually uses:

- DPLL + learning from conflicts
- When you hit a dead end, analyze *why*
- Add learned clauses to prevent revisiting mistakes
- Vastly faster than vanilla DPLL

Key insight: Z3 is essentially running CDCL in the background. Your constraints get translated into a boolean formula, then solved using highly optimized SAT-solving techniques.